

Exercise # 14.1

Double integrals Questions

5-12 Evaluate the iterated integrals.

$$5 \quad \int_0^{\ln 3} \int_0^{\ln 2} dy dx \quad \because \int e^y dy = e^y$$

$$\int_0^{\ln 3} (e^x \cdot e^y)_{y=0}^{y=\ln 2} dx \quad \int e^x dx = e^x$$

$$\begin{aligned} \int_0^{\ln 3} e^x [e^{\ln 2} - e^0] dx &= \int_0^{\ln 3} e^x [2 - 1] dx \\ &= e^{\ln 3} - e^0 = [e^x]_{x=0}^{x=\ln 3} \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

$$6: \int_0^2 \int_0^1 y \sin x dy dx$$

$$= \int_0^2 \left[y^2 \sin x \right]_{y=0}^{y=1} dx = \int_0^2 \sin x \left(\frac{1^2}{2} - \frac{0^2}{2} \right) dx$$

$$= \int_0^2 \sin x \left(\frac{1}{2} \right) dx = \frac{1}{2} \int_0^2 \sin x dx$$

$$= \frac{1}{2} (-\cos x) \Big|_{x=0}^{x=2} = -\frac{1}{2} [\cos 2 - \cos 0]$$

$$= -\frac{1}{2} (\cos 2 - 1)$$

$$= \frac{1 - \cos 2}{2}$$

$$7: \int_{-1}^0 \int_2^5 dx dy \quad \because \int dx = x$$

$$\int_{-1}^0 [x]_2^5 dy = \int_{-1}^0 (5-2) dy = \int_{-1}^0 3 dy$$

$$3 \int_{-1}^0 dy = [3y]_{-1}^0 = [3(0) - 3(-1)] = 3$$

$$8: \int_4^6 \int_{-3}^7 dy dx = \int_4^6 [y]_{-3}^7 dx$$

$$= \int_4^6 (7 - (-3)) dx = \int_4^6 10 dx$$

$$= [10x]_4^6 = 10(6) - 10(4) = 20$$

$$9: \int_0^1 \int_0^1 \frac{x}{(xy+1)^2} dy dx$$

$$\int_0^1 \int_0^1 (xy+1)^{-2} x dy dx$$

$$\therefore \int (fy)^n f'(y) dy = \frac{f(y)^{n+1}}{n+1}$$

$$= \int_0^1 \left[(xy+1)^{-2+1} \right]_{y=0}^{y=1} dx$$

$$= \int_0^1 \left[-1(2y+1)^{-1} \right]_{y=0}^{y=1} dx$$

$$= \int_0^1 \left[\left(\frac{-1}{xy+1} \right) \right]_{y=0}^{y=1} dx$$

$$= \int_0^1 \left[\frac{-1}{x(1)+1} - \frac{-1}{x(0)+1} \right] dx$$

$$= \int_0^1 \left(\frac{-1}{x+1} + 1 \right) dx$$

$$= - \int_0^1 \left(\frac{1}{x+1} - 1 \right) dx = - \int \frac{1}{x+1} + \int dx$$

$$= - \left[\ln(x+1) - x \right]_{x=0}^{x=1}$$

$$= - \left[\ln(1+1) - 1 - (\ln(0+1) - 0) \right]$$

$$= - \left[\ln 2 - 1 - 0 \right]$$

$$= 1 - \ln 2$$

$$10: \int_{\pi/2}^{\pi} \int_1^2 x \cos xy \, dy \, dx$$

$$\int_{\pi/2}^{\pi} x \left[\frac{(\sin xy)}{x} \right]_{y=1}^{y=2} dx$$

$$\int_{\pi/2}^{\pi} [\sin 2x - \sin x] dx$$

$$= \left[-\frac{\cos 2x}{2} + \cos x \right]_{\pi/2}^{\pi}$$

$$= \left(\left[-\frac{\cos 2\pi}{2} + \cos \pi \right] - \left[-\frac{\cos 2\frac{\pi}{2}}{2} + \cos \frac{\pi}{2} \right] \right)$$

$$= \left(\left[-\frac{(1)}{2} - 1 \right] - \left[-\frac{(1)}{2} + 0 \right] \right)$$

$$= -\frac{1}{2} - 1 - \frac{1}{2}$$

$$= -2$$

$$11: \int_0^{\ln 2} \int_0^1 xy e^{y^2 x} \, dy \, dx$$

$$\int_0^{\ln 2} \frac{1}{2} \int_0^1 e^{xy^2} xy \, dy \, dx = \frac{1}{2} \int_0^{\ln 2} \left[e^{xy^2} \right]_{y=0}^{y=1} dx$$

$$= \frac{1}{2} \int_0^{\ln 2} [e^{x(1)^2} - e^{x(0)^2}] dx$$

$$= \frac{1}{2} \int_0^{\ln 2} (e^x - 1) dx$$

$$= \frac{1}{2} \left[e^x - x \right]_{x=0}^{x=\ln 2}$$

$$= \frac{1}{2} \left[(e^{\ln 2} - \ln 2) - (e^0 - 0) \right]$$

$$= \frac{1}{2} [2 - \ln 2 - 1]$$

$$= \frac{1 - \ln 2}{2}$$

$$12: \int_3^4 \int_1^2 \frac{1}{(x+y)^2} dy dx$$

$$= \int_3^4 \int_1^2 (x+y)^{-2} dy dx = \int_3^4 \left[\frac{(x+y)^{-2+1}}{-2+1} \right]_{y=1}^{y=2} dx$$

$$= \int_3^4 \left[\frac{-1}{(x+y)} \right]_{y=1}^{y=2} dx$$

$$= \int_3^4 \left[\frac{1}{x+2} - \frac{1}{x+1} \right] dx$$

$$= - \left[\ln(x+2) - \ln(x+1) \right]_3^4$$

$$= - \left[(\ln(4+2) - \ln(4+1)) - (\ln(3+2) - \ln(3+1)) \right]$$

$$\therefore \ln x + \ln y = \ln xy$$

$$\therefore \ln a - \ln b = \ln \frac{a}{b}$$

$$= -[\ln 6 - \ln 5 - \ln 5 + \ln 4]$$

$$= -[\ln 6 \times 4 - \ln 5 \times 5]$$

$$= -[\ln 24 + \ln 25] = -\ln 24 + \ln 25$$

$$= \ln\left(\frac{25}{24}\right)$$

13-15 Evaluate the double integral over the rectangular region R.

$$13: \int_R \int 4xy^3 dA, R = \{(x, y) : -1 \leq x \leq 1, -2 \leq y \leq 2\}$$

$$= \int_{-1}^1 \int_{-2}^2 4xy^3 dy dx = \int_{-1}^1 4x \left[\frac{y^4}{4} \right]_{y=-2}^{y=2} dx$$

$$= \int_{-1}^1 x [(2)^4 - (-2)^4] dx = \int_{-1}^1 x (16 - 16) dx$$

$$= \int_{-1}^1 0 dx = 0$$

$$14: \int_R \int \frac{xy}{\sqrt{x^2+y^2+1}} dA;$$

$$R = \{(x,y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$$

$$= \int_0^1 \int_0^1 \frac{x}{2} (x^2+y^2+1)^{-1/2} 2y dy dx$$

$$= \int_0^1 \frac{x}{2} \left[\frac{(x^2+y^2+1)^{1/2+1}}{-1/2+1} \right]_{y=0}^{y=1} dx$$

$$= \int_0^1 \frac{x}{2} \left(\sqrt{x^2+y^2+1} \right)_{y=0}^{y=1} dx$$

$$= \int_0^1 \frac{x}{2} \left[\sqrt{x^2+(1)^2+1} - \sqrt{x^2+0+1} \right] dx$$

$$= \int_0^1 \frac{x}{2} \left(\sqrt{x^2+y^2+1} \right)_{y=0}^{y=1} dx$$

$$= \int_0^1 x \left[\sqrt{x^2+(1)^2+1} - \sqrt{x^2+0+1} \right] dx$$

$$= \int_0^1 x \left(\sqrt{x^2+2} - \sqrt{x^2+1} \right) dx$$

$$= \int_0^1 \left[(x^2+2)^{1/2} \frac{2x}{2} dx - (x^2+1)^{1/2} \left(\frac{2x}{2} \right) dx \right]$$

$$= \frac{1}{2} \left[\frac{(x^2+2)^{3/2}}{3/2} - \frac{(x^2+1)^{3/2}}{3/2} \right]_0^1$$

$$= \frac{1}{3} \left[(x^2+2)^{3/2} - (x^2+1)^{3/2} \right]_0^1$$

$$= \frac{1}{3} \left((1+2)^{3/2} - (1+1)^{3/2} - \left((2)^{3/2} - 1 \right) \right)$$

$$= \frac{1}{3} \left(\sqrt{3^3-2} - \sqrt{2^3+1} \right) \text{ or } \frac{1}{3} [3\sqrt{3} - 4\sqrt{2} + 1]$$

15: $\int_R \int x \sqrt{1-x^2} dA : R = \{(x,y) : 0 \leq x \leq 1, 2 \leq y \leq 3\}$

$$\int_0^1 \int_2^3 x \sqrt{1-x^2} dy dx = \int_0^1 x \sqrt{1-x^2} (y)_{y=2}^{y=3} dx$$

$$= \int_0^1 x \sqrt{1-x^2} (3-2) dx = \int_0^1 (1-x^2)^{1/2} x dx$$

$$= -\frac{1}{2} \int_0^1 (1-u^2)^{1/2} (-2u) du = -\frac{1}{2} \left[\frac{(1-x^2)^{1/2+1}}{1/2+1} \right]_0^1$$

$$= \frac{1}{2} \times \frac{2}{3} \left[(1-x^2)^{3/2} \right]_0^1$$

$$= -\frac{1}{3} \left[(1-(1)^2)^{3/2} - (1-0) \right] = -\frac{1}{3} [0-1] = \frac{1}{3}$$

$$16: \iint_R (x \sin y - y \sin x) dA;$$

$$R = \{(x, y) : 0 \leq x \leq \pi/2, 0 \leq y \leq \pi/3\}$$

$$= \int_0^{\pi/2} \int_0^{\pi/3} (x \sin y - y \sin x) dy dx$$

$$= \int_0^{\pi/2} \left[-x \cos y - \sin x \frac{y^2}{2} \right]_{y=0}^{y=\pi/3} dx$$

$$= \int_0^{\pi/2} \left[\left(-x \cos \frac{\pi}{3} - \frac{\sin x}{2} \left(\frac{\pi}{3} \right)^2 \right) - \left(-x \cos 0 - \frac{\sin x}{2} (0)^2 \right) \right] dx$$

$$= \int_0^{\pi/2} \left[-\frac{x}{2} - \frac{\pi^2}{18} \sin x + x \right] dx = \int_0^{\pi/2} \left(\frac{x}{2} - \frac{\pi^2}{18} \sin x \right) dx$$

$$= \left[\frac{x^2}{4} - \frac{\pi^2}{18} (-\cos x) \right]_0^{\pi/2}$$

$$= \left(\left(\frac{\pi}{2} \right)^2 + \frac{(\pi)^2}{18} \cos \frac{\pi}{2} \right) - \left(\frac{(0)^2}{4} + \frac{\pi^2}{18} \cos 0 \right)$$

$$= \frac{\pi^2}{16} + 0 - 0 - \frac{\pi^2}{18}$$

$$= \frac{9\pi^2 - 8\pi^2}{144} = \frac{\pi^2}{144}$$